LECTURE ACTIVITY 14: TAGGING PENGUINS

STAT 1110: Applied Statistical Concepts and Methods – Fall 2026 with Dr. Ethan P. Marzban

This activity is heavily adapted from one created by the STAT 1110 Course Coordination Team, which in turn is modified from "Introduction to Statistical Investigations," by Tintle et. al.

Learning Objectives:

By the end of this activity, you should be able to

1) State the null and the alternative hypotheses in terms of "no association" versus "there is an association" as well as in terms of comparing population proportion of success for two categories of the explanatory variable (i.e., $\pi 1$ and $\pi 2$) when exploring the relationship between two categorical variables.
2) Implement the 3S strategy: find a statistic, simulate, and compute the strength of evidence against observed study results happening by chance alone.
3) Describe how to use cards to simulate what outcomes (in terms of difference in conditional proportions and/or relative risk) are to be expected in repeated random assignments, if there is no association between the two variables.
4) Conduct a simulation-based analysis when evaluating claims about two population proportions.
5) Find and interpret the p-value for a test of two proportions.
6) State a complete conclusion about the alternative hypothesis (and null hypothesis) based on the magnitude of the standardized statistic and/or p-value.
7) Identify whether the validity conditions for a two-sample z-test are met.
8) Conduct a theory-based approach when evaluating claims about two population proportions.
9) Use the Theory-based method to find a 95% confidence interval for the difference in population proportions of success for two groups, and interpret the interval in the context of the study.

Key Terms:

- 3S Method
- Two-Sample Z-Test
- Confidence Interval

Introduction

Are metal bands used for tagging penguins possibly affecting them in some way? Researchers[1] investigated this question with a sample of 20 penguins near Antarctica. All of these penguins had already been tagged with RFID chips, and the researchers randomly assigned 10 of them to receive a metal band on their flippers in addition to the RFID chip. The other 10 penguins did not receive a metal band. Researchers then kept track of which penguins survived for the 4.5-year study and which did not.

[1] Saraux, C., Le Bohec, C., Durant, J. et al. Reliability of flipper-banded penguins as indicators of climate change. *Nature* 469, 203–206 (2011). https://doi.org/10.1038/nature09630

Hypothesis Testing

Question 1

(a) Identify the observational unit in this study.

(b) Identify the two variables in this study, and classify them as either categorical or quantitative.

(c) Identify which of the two variables is the explanatory variable, and which is the response.

- **Explanatory Variable:** _____
- **Response Variable:** _____

Question 2

Why did the researchers include a control group in this study? Why didn't they just see how many penguins survived while wearing a metal band?

Question 3

Is this an observational study, an experiment, or a randomized experiment? Explain how you know.

- ☐ Observational study
- ☐ Experiment
- ☐ Randomized experiment

Justification:

The two competing hypotheses to be tested can be phrased as:

- **Hypothesis 1:** There is no association between whether the penguin gets the metal band and whether the penguins survive.
  - Equivalently: Metal bands have no effect on penguins' survival.
- **Hypothesis 2:** There is an association between whether the penguin gets the metal band and whether the penguins survive.
  - Equivalently: Metal bands have an effect in penguins' survival. survival.

Question 4

Which hypothesis is the null hypothesis, and which is the alternative hypothesis?

- **Hypothesis 1**: ☐ Null Hypothesis ☐ Alternative Hypothesis
- **Hypothesis 2**: ☐ Null Hypothesis ☐ Alternative Hypothesis

The hypotheses as stated before are in terms of "association," which is a little difficult to quantify. Let's see if we can make things more formal, by considering **parameters**. Notice that we have two groups, leading to *two* natural parameters to consider:

- π_{Control}, the population proportion of all untagged penguins that survived the course of the 4.5-year time period of observation
- π_{MB}, the population proportion of all tagged penguins that survived the course of the 4.5-year time period of observation

The null and alternative hypotheses can then be rephrased as follows:

- H_0: The population proportion of penguins who would survive when they do **not** receive a metal band is the same as when they do.
  - That is; $H_0 : \pi_{\text{Control}} = \pi_{\text{MB}}$
- H_A: The population proportion of penguins who would survive when they do **not** receive the metal band is different than when they do.
  - That is; $H_A : \pi_{\text{Control}} \neq \pi_{\text{MB}}$

In the last lecture activity, we saw a few different ways to "combine" quantities. One of those is by taking differences - something that seems appealing in this context. That is, note that we can equivalently phrase our hypotheses as:

$$\begin{bmatrix} H_0 : & \pi_{\text{Control}} - \pi_{\text{MB}} = 0 \\ H_A : & \pi_{\text{Control}} - \pi_{\text{MB}} \neq 0 \end{bmatrix}$$

So, in essence, we can think of $(\pi_{\text{Control}} - \pi_{\text{MB}})$ as a single quantity of interest, about which we can phrase our hypotheses.

Now for the data! The researchers found that 9 of 20 penguins survived, of whom 3 had a metal band and 6 did not.

Question 5

Fill in the missing counts in the following two-way table. **Hint:** Start by entering 40 and 34 into the appropriate cells. Then make sure that both rows and both columns add up to the indicated totals.

	No Metal Band (control)	Metal Band (treatment)	Total
Survived			
Did Not Survive			
Total	**10**	**10**	**20**

Question 6

Calculate the sample proportion of survivors in each group. Also calculate the difference between these proportions (subtracting the "metal band" proportion from the "control" proportion).

- Sample proportion of survivors in control group: $\hat{p}_{\text{Control}} =$ _____
- Sample proportion of survivors in treatment group: $\hat{p}_{\text{MB}} =$ _____
- Difference in sample proportions of survivors (control − MB): $\hat{p}_{\text{Control}} - \hat{p}_{\text{MB}} =$ _____

Scratch Work:

Question 7

Why is the difference in sample proportions you calculated in Question (6) positive?

Question 8

Is it *possible* that this difference could have happened even if the metal band had no effect; i.e., simply due simply to the random nature of assigning penguins to groups?

Simulation-Based Approach

We know it is *possible* that the difference in sample proportions of 0.30 could have happened even if the metal band had no effect, but now we need to figure out how *likely* (or *unlikely*) it is to have happened by chance alone. That is - though the proportion of penguins in the METAL BAND group who survived was greater than the proportion who survived in the control group (NO METAL BAND), we don't yet know if this provides convincing evidence of a genuine difference in the population proportions.

Recall that this is precisely where the **3S** method can help us! We've taken care of the first "S," **statistic** - now it's time to **simulate** (and then quantify strength of evidence). In particular, we'll apply the same logic that we have used previously: to use a simulation analysis to determine whether the observed results are typical or surprising for what we would find if there is no effect from the metal band (i.e if the the null hypothesis were true).

As with earlier studies, there are two possible explanations for the results: the difference occurred by chance (there is no effect of metal band) or the difference did not occur by chance (metal band affects penguins).

The key to our simulation analysis is to assume that if there is no effect of metal band (null hypothesis), then **the 9 penguins would have survived regardless of whether or not they had received the metal band**. Similarly, we'll assume that the 11 non-surviving penguins would not have survived, no matter which group they had been assigned. In other words, our simulation assumes the null hypothesis is true – that there is no association, no connection, between receiving the metal band and survival.

Concept Overview

To evaluate the significance of the observed difference in our groups, we will investigate how large the difference in sample proportions of each group tends to be just from the random assignment of response outcomes to the explanatory variable groups.

We cannot use coins to conduct this simulation analysis, because we have *two* variables to consider: whether penguins received the metal band, and whether the penguin survived. Instead of coins, we will use index cards. Here's our strategy:

1) Take a set of _____ cards, with _____ blue cards representing the survivors and _____ red cards representing those who did not survive.

2) Shuffle the cards well, and randomly deal out _____ to be the metal band group and the remaining _____ go to the control group.

3) Count how many survivors (blue cards) you have in each group and how many non-survivors (red cards) you have in each group.

4) Construct the two-way table to show the number of survivors and non-survivors in each group. (Clearly nothing different happened to those in the control group and those in metal band group – any differences between the two groups that arise are due purely to the random assignment process.)

Question 9

(a) What does one card represent in this simulation?

(b) What is the total number of cards we should use? How many should be blue, and how many should be red?

(c) In step 2, how many cards should we draw and assign to the metal band group? How many cards should we draw and assign to the control group?

Question 10

In pairs, carry out this procedure (steps 1 through 4) twice, letting each person get one opportunity to shuffle and draw. Record your results in the following two-way tables each time, and calculate the required quantities.

	No Metal Band (control)	**Metal Band (treatment)**	**Total**
Survived			
Did Not Survive			
Total			

- Sample proportion of survivors in control group: $\hat{p}_{\text{Control}} =$ _____
- Sample proportion of survivors in treatment group: $\hat{p}_{\text{MB}} =$ _____
- Difference in sample proportions of survivors (control − MB): $\hat{p}_{\text{Control}} - \hat{p}_{\text{MB}} =$ _____

	No Metal Band (control)	**Metal Band (treatment)**	**Total**
Survived			
Did Not Survive			
Total			

- Sample proportion of survivors in control group: $\hat{p}_{\text{Control}} =$ _____
- Sample proportion of survivors in treatment group: $\hat{p}_{\text{MB}} =$ _____
- Difference in sample proportions of survivors (control − MB): $\hat{p}_{\text{Control}} - \hat{p}_{\text{MB}} =$ _____

Question 11

Come to the front of the room, and add a dot on our collective dotplot indicating your two differences in proportions.

Of course, to effectively carry out a simulation-based procedure to test our hypotheses we need a *lot* of repetitions. This is where technology comes to our rescue (yet again!).

1) Navigate to the Two Proportions Applet.
2) Click the button next to "**Enter Table 2 x 2**."
3) Enter the original observed counts (see Question (5 above) in the 2×2 table for this study into the table in the upper left cell of the applet.
    - Instead of "GroupA" type "Control"
    - Instead of "GroupB" type "MetalBand" (make sure there is no space)
    - Instead of "Success" type "Survived"
    - Instead of "Failure" type "NotSurvived" (make sure there is no space)
    - Type the numbers of survivals and deaths for each group
4) Press the "**Use Table**" button. Notice that the applet produces a segmented bar graph and calculates the observed difference in success proportions.
5) Check the "**Show Shuffle Options**"
6) Under Select Display, check **Cards**
7) Press the Shuffle button.

Notice that the applet has done exactly what we did in Question (10) above! The applet also determines the 2×2 table for the simulated results and plots the shuffled difference in sample proportions for each group on the graph to the right.

Question 12

What is the value of the first simulated difference in proportions of success?

Now increase the **Number of Shuffles** to 9,999 and press the **Shuffle** button for the applet to generate a total of 10,000 random assignments.

Question 13

Using the context of the problem, explain what a dot on the plot represents.

Question 14

You should have noticed that the center of the distribution is about 0. Why is this the case?

As you know, you can assess the strength of evidence against the null hypothesis by calculating a p-value. In this example, to calculate a p-value, you use the null distribution and count the proportion of simulated samples that produced a difference in proportions equal to the original observed statistic ($\hat{p}_{\text{control}} - \hat{p}_{\text{MB}} = 0.3$) or more extreme values in **both directions**. We need to consider both directions because we are conducting a two-sided test with $H_A : \pi_{\text{Control}} - \pi_{\text{MB}} \neq 0$.

Question 15

Calculate the *p*-value. To conduct a two-sided test, make sure you click on drop-down menu next to "count sample" and select "Beyond". Paste a copy of the plot showing the calculated *p*-value. **To show your work, include a screenshot of the entire screen showing the inputs (anything you typed in the applet) and outputs including (1) null distribution, (2) observed result and (3) p-value.**

- My simulation-based approximate *p*-value is _____

Here is my screenshot:

Question 16

Interpret this *p*-value. (By now, we expect you to know what this means.)

Question 17

Evaluate this *p*-value. How much evidence does the sample data provide against the null hypothesis?

- ☐ Not much evidence against the null hypothesis
- ☐ Moderate evidence against the null hypothesis
- ☐ Strong evidence against the null hypothesis
- ☐ Very strong evidence against the null hypothesis

I have a confession to make: The results described above were actually just a small part of the study! We chose to start with partial results to make the by-hand simulation with cards fairly manageable. But the actual study involved 100 penguins near Antarctica, with 50 randomly assigned to each group. The researchers found that 16 of 50 survived in the metal band group, and 31 of 50 survived in the control group. The actual results from the full study in the 2×2 table are shown below:

	No Metal Band (control)	**Metal Band (treatment)**	**Total**
Survived	31	16	**47**
Did Not Survive	19	34	**53**
Total	**50**	**50**	**100**

Table 1: Two-way table of the full dataset.

Question 18

Determine the conditional proportions who survived in each group. Also calculate the difference in these proportions (control – metal band). Is the value of this difference similar to what you calculated earlier for the partial study?

- Sample proportion of survivors in control group: $\hat{p}_{\text{Control}} =$ _____
- Sample proportion of survivors in treatment group: $\hat{p}_{\text{MB}} =$ _____
- Difference in sample proportions of survivors (control − MB): $\hat{p}_{\text{Control}} - \hat{p}_{\text{MB}} =$ _____

Question 19

(a) Navigate back to the Two Proportion Applet, and update the table to use the full data (displayed in Table (1) above). Use at least 10,000 repetitions to obtain a new simulation-based approximate *p*-value.

- My simulation-based approximate *p*-value is _____

(b) ***Evaluate*** this *p*-value. How much evidence does the sample data provide against the null hypothesis?

- ◯ Not much evidence against the null hypothesis
- ◯ Moderate evidence against the null hypothesis
- ◯ Strong evidence against the null hypothesis
- ◯ Very strong evidence against the null hypothesis

Question 20

Summarize the conclusion that you draw from this study based on your analysis. Make sure you explain the reasoning process behind your conclusion by referring to the (1) observed statistic, (2) p-value (3) justify strength of evidence against the null model (very strong, strong, moderate, weak) based on the p-value, (4) reject/fail to reject the null hypothesis, and (5) overall conclusion in context. Make sure you refer to the parameter in your conclusion.

Question 21

As a reminder, the sample in this study was composed of 100 king penguins on the Crozet Islands. Do you think the results of this study apply to all penguins? Justify your answer.

Question 22

Is drawing a cause-and-effect conclusion (between the metal bands and a lower chance of penguin survival) justified from this study? Explain why or why not. (Be sure to identify both reasons for the correct conclusion.)

Question 23

As we discussed in Chapter 2, in every statistical analysis it is possible to make a Type I or a Type II error. Identify which error is possible here and describe it in context.

Theory-Based Approach

Just as we learned in Chapter 1 about a **theory-based approach** for making inferences about a single long-run/population proportion (**one-sample z-test**), now you will learn a theory-based approach for making inferences about comparing long-run/population proportions between two groups in the population (**two-sample z-test**). This theory-based approach is again based on an approximation from a **normal distribution**.

We'll again use the Theory-Based Inference Applet to conduct our theory-based analyses.

1) Go to the Theory-Based Inference Applet.
2) Use the pull-down menu to select **Two proportions**.
3) Enter the relevant summary information, using the control sample as group 1.
4) Press **Calculate**.
5) Check the **Test of Significance** box.
6) Press on the inequality sign button until the alternative hypothesis becomes two-sided.
7) Press **Calculate**.

Question 24

Report the t**heory-based *p*-value (two sample z-test)** using the applet. To show your work, include a screenshot of the entire screen showing the inputs (anything you typed in the applet) and outputs including (1) the data you input for each group, (2) statistic/bar graph, (3) null distribution, and (4) theory-based p-value.

- My theory-based approximate *p*-value is _____

Here is my screenshot:

Question 25

What is the value of the test-statistic (or standardized statistic)? Interpret this value.

As with a single proportion, the theory-based inference approach for comparing two proportions is only valid when a normal distribution provides a good approximation to the distribution of the statistic $\hat{p}_1 - \hat{p}_2$. Conservatively, the theory-based significance test and confidence interval work well when there are at least 10 observations in each of the four cells of the two-way table. In other words, we require at least 10 "successes" and at least 10 "failures" in each sample in order to use this theory-based approach.

Concept Overview

Validity Conditions: The theory-based test and interval for the difference in two proportions (called a two-sample z-test or two-sample z-interval) work well when there are at least 10 observations in each of the four cells of the 2×2 table.

Question 26

Is it valid to use the theory-based approach for two proportions in this study? Explain.

Estimation

At this point we know that the results of the study are statistically significant[2] and there is a real difference between the population proportion of penguins that survive when they do not receive a metal band and when they do. But now we should be wondering: how big is this difference?

Remember that our general working formula for a confidence interval is

$$(\textbf{statistic}) \pm (\textbf{multiplier}) \cdot (\textbf{standard error})$$

Here, our parameter of interest is actually a difference of population proportions $(\pi_{\text{Control}} - \pi_{\text{MB}})$; as such, a natural point estimator is the difference of *sample* proportions $(\hat{p}_{\text{Control}} - \hat{p}_{\text{MB}})$, leading us to a confidence interval of the form

$$(\hat{p}_{\text{Control}} - \hat{p}_{\text{MB}}) \pm \text{SE}(\hat{p}_{\text{Control}} - \hat{p}_{\text{MB}})$$

As far as the standard error is concerned, we have a couple of options available to us:

- Using SD_{NULL} from a simulation
- Using the following formula:

$$\text{SE}(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(2-\hat{p}_2)}{n_2}}$$

  where:

  - $\hat{p}_1$ and $\hat{p}_2$ denote the sample proportion of successes in groups 1 and 2, respectively
  - n_1 and n_2 denote the sample sizes of groups 1 and 2, respectively

[2] Even though the textbook uses this terminology, I've been a bit careful to avoid using it until now. Basically, "statistically significant" just means what it says - the difference seems unlikely to be due solely to chance.

Question 27

(a) Calculate $\text{SE}(\hat{p}_1 - \hat{p}_2)$ using the data from Table (1) above.

(b) Construct an approximate 95% confidence interval for $(\pi_{\text{Control}} - \pi_{\text{MB}})$. **Hint:** remember that the appropriate multiplier for a 95% z-interval is around 2.

(c) Navigate back to the Theory-Based Inference Applet and compute an approximate 95% confidence interval for $(\pi_{\text{Control}} - \pi_{\text{MB}})$. Check that it is similar to the one you calculated in Part (b) above.

(d) Provide an interpretation of the confidence interval. Use the skeleton below to help.

> We are 95% confident that the population proportion of penguins who would survive when they _____ receive a metal band is between 11 to 49 percentage points _____ than when they _____ receive a metal band, for all king penguins similar to the ones in the study.

In the skeleton interpretation above, there are very important details that we want you to pay attention to:

- Referring to the population proportions:
  - π_{Control} (population proportion of penguins who would survive without the metal band) and
  - π_{MB} (population proportion of penguins who would survive with metal band)
- Identifying which population proportion was larger
- Referring to the population: king penguins near Antarctica similar to the ones in the study.

Question 28

Why do you think zero was NOT included in the 95% confidence interval?